

Four Majid Indoor-Localization Protocols: A Fair, Executable Reconstruction and Method Benchmark

Integrated technical report

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Abstract

This report integrates four paper-specific tasks rather than pooling unlike experiments: MPURge–MAP receiver localization, original MPURge virtual- transmitter (VT) construction, SMART–LEA refinement, and the two-room CNN fingerprinting task. Every numerical result is labelled either a completed replacement-scene benchmark, a diagnostic smoke result, published context, or pending evidence. Fairness is the paper-specific number and location of physical acquisitions, not a delay-only rule: noisy power, AoA, and complex CIR are legal when obtained from those same acquisitions. Hidden ray/VT IDs, query truth, clean side channels, and extra spatial supervision are forbidden. The report gives a beginner-reproducible recipe for every method appearing in a table, complete leaderboards for finished protocols, explicit hooks for the two pending full runs, and a release/audit ledger.

Release state. Corrected MPURge–MAP and SMART–LEA full runs are complete and audited. The original-MPURge table is a one-scene, two-noise diagnostic smoke result only; its 24-block full attempt did not write aggregate outputs. The corrected CNN full result is pending and the older smoke metrics are superseded. No missing table is silently filled with an earlier or incompatible experiment.

Contents

1	What may and may not be claimed	2
1.1	The four distinct protocols	2
1.2	Shared acquisition and metric contract	2
2	Protocol I: corrected MPURge–MAP	3
2.1	Frozen reconstruction	3
2.2	Common equations	3
2.3	Recipe index for all 25 rows	4
2.4	Complete pooled leaderboard	5
3	Protocol II: original MPURge VT construction	5
3.1	Corrected task and evidence boundary	5
3.2	Construction and scoring equations	5
3.3	Recipe index for every diagnostic row	6
3.4	Complete smoke leaderboard: diagnostic, not final	7
4	Protocol III: SMART–LEA	7
4.1	Frozen full reconstruction	7
4.2	Core equations and method recipes	7
4.3	Complete SMART macro leaderboard	8
4.4	All 30 methods in all eight SMART cells	9

5	Protocol IV: two-room CNN fingerprinting	11
5.1	Printed protocol and corrected reconstruction	11
5.2	Equations and the 25-method recipe index	11
6	Compact causal and multilevel evidence	13
6.1	Confirmed positive causal sequential result	13
6.2	Honest causal multilevel calibration: useful, not a confirmed win	13
7	Release and audit ledger	14
7.1	Hash and provenance resolution	14
7.2	Exact evidence paths	14
7.3	Minimal reproducibility checklist	15
8	Bottom line	15

1 What may and may not be claimed

The four source papers use simulator projects, scene files, and random seeds that are not available in this workspace. Consequently, a number produced here is an *executable replacement-scene reconstruction*; it is not an exact reproduction of the paper’s number and not an external state-of-the-art claim. Published values appear only as context and are never pooled with reconstruction rows.

The paper/reconstruction boundary is operational:

1. A printed paper fact is stated as such and tied to its protocol.
2. A missing choice—geometry, AP coordinate, corruption extractor, threshold, seed, or ambiguity—is disclosed as a reconstruction choice.
3. A protected query may score a frozen method but may not select a model, threshold, sensor channel, fallback, or ablation.
4. A method may use every noisy physical observable at the same permitted acquisitions. It may not add labelled positions or read latent simulator identities, clean geometry, the true obstacle label, or query coordinates.

1.1 The four distinct protocols

Protocol	Output	Paper-side budget	Replacement implementation	Evidence state
MPUrge-MAP	One static query coordinate	Uniform 0.5 m RP map; 100 non-RP queries; independent range AWGN and obstruction tests	$4 \times 5 \times 3$ m image-source room; 80 cell-centred RPs; 100 locked off-grid queries; 16 conditions	Full, audited
Original MPUrge	Unordered VT coordinate set per transmitter	Nearby surveyed RPs; four physical transmitters and 16 VTs; several noise realizations; exact scene/layout unavailable	Disclosed 20×20 m variable-visibility scenes; same sparse survey/query acquisition rule; corrected $P = 5$ and matching brackets	Smoke only; full pending
SMART-LEA	One static coordinate refined from an initial pose and VT/Tx anchors	Irregular floorplan; four physical plus 16 virtual transmitters; spacings 0.25, 0.5, 1.5, 2.5 m	25×25 m rectangle; four Tx and four wall images/Tx; 1,000 locked queries; 3 seeds; clean/noisy	Full, audited
CNN	One static coordinate or RP posterior	Per room: 20 RPs, 60 AP positions, 7 configurations (8,400 fingerprints); 1,600 off-grid queries	Same counts and room sizes; image-source/noisy-array receiver; corrected grouped split and leave-self map logic	Corrected full pending

Table 1: The tasks have different outputs and supervision budgets; their errors are not one common leaderboard.

1.2 Shared acquisition and metric contract

Latent paths first synthesize a noisy array impulse response,

$$y_a[n] = \sum_{\ell} \alpha_{\ell} e^{-j(2\pi/\lambda) \mathbf{p}_a^{\top} \mathbf{u}_{\ell}} \mathbf{1}[n = n_{\ell}] + w_a[n], \quad w_a[n] \sim \mathcal{CN}(0, \sigma^2). \quad (1)$$

Peak extraction yields a token $z_{\ell} = (\hat{r}_{\ell}, \hat{P}_{\ell}, \hat{\mathbf{u}}_{\ell})$ and leaves the noisy complex CIR available. Missing detection removes the whole token, so delay, power, AoA, dropout, and CIR noise are coupled. Survey and query acquisitions use independent draws. Latent source/ray identity and reflection order are never estimator features.

For a query truth $\mathbf{x}_b$ and prediction $\hat{\mathbf{x}}_b$,

$$e_b = \|\hat{\mathbf{x}}_b - \mathbf{x}_b\|_2, \quad \bar{e} = B^{-1} \sum_b e_b, \quad \text{RMSE} = \sqrt{B^{-1} \sum_b e_b^2}. \quad (2)$$

Median and 90th percentile are also retained. Pairwise uncertainty uses the same physical blocks for both methods: with $\Delta_b = e_b^{\text{baseline}} - e_b^{\text{candidate}}$, a positive Δ means the candidate is better. Blocks, not augmentation duplicates, are resampled.

2 Protocol I: corrected MPUrgE–MAP

2.1 Frozen reconstruction

Every method receives the same 80 survey locations and the same 100 locked off-grid query locations. The 16 cells are ideal, nine range-AWGN levels $\sigma_r \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5\}$ m, five individual objects, and all objects. Each cell has one materialised survey tensor and an independently corrupted query tensor shared verbatim across methods. Learned augmentation corrupts the 80 stored fingerprints only (1,280 rows; zero new simulator coordinates). Selection uses 64 whole RP groups for fit and 16 for validation. Seven neural methods use seeds 211, 337, and 461. The 100 query labels are opened only for final scoring.

The observable channels are delay, noisy received power, noisy receive AoA, and noisy 8-element CIR. Zero-range-noise cells use the frozen high-SNR receiver setting (80 dB); range-AWGN cells use 25 dB for other modalities. That sensitivity choice is part of the replacement protocol and is a reason not to extrapolate its multimodal margin directly to hardware.

2.2 Common equations

For sorted query set $Q = \{q_a\}$ and RP set $R_i = \{r_{ib}\}$, corrected MPUrgE uses

$$\delta_{ab} = \alpha|q_a - r_{ib}| + (1 - \alpha)d_{\text{pat}}(a, b), \quad (3)$$

$$d_{\text{pat}}(a, b) = \binom{2p+1}{2}^{-1} \sum_{u < v} |W_a(u) - W_a(v)| - |W_{ib}(u) - W_{ib}(v)|, \quad (4)$$

$$C_i = \frac{|\mathcal{M}_i|}{\max(|Q|, |R_i|, 1)}, \quad s_i = \frac{1}{C_i |\mathcal{M}_i|} \sum_{(a,b) \in \mathcal{M}_i} \delta_{ab}. \quad (5)$$

Mutual-nearest matches are processed in increasing composite δ , and a new match may not reverse the delay order of an accepted match. The baseline uses $p = 6$, $\alpha = 0.7$, normalized pattern distance, and inverse-score interpolation over the three best finite RPs:

$$w_i = \frac{s_i^{-1}}{\sum_{j \in K} s_j^{-1}}, \quad \hat{\mathbf{x}} = \sum_{i \in K} w_i \mathbf{x}_i. \quad (6)$$

No finite evidence returns the survey-coordinate centroid for every method.

Three reusable decoders are

$$D_i^{\text{Ch}} = \frac{1}{2} \left[|Q|^{-1} \sum_{q \in Q} \min_{r \in R_i} |q - r| + |R_i|^{-1} \sum_{r \in R_i} \min_{q \in Q} |r - q| \right], \quad (7)$$

$$w_i^{(T)} = \frac{\exp[-(s_i - s_{(1)})/T]}{\sum_{j \in K} \exp[-(s_j - s_{(1)})/T]}, \quad (8)$$

$$p^{(t+1)} = (1 - \gamma)p^{(t)} + \gamma G^\top p^{(t)}, \quad \hat{\mathbf{x}} = \sum_i p_i^{(L)} \mathbf{x}_i, \quad (9)$$

where G is the normalized six-neighbour RP graph. The frozen EvoMDP row uses $E_i = \sum_{k=1}^8 w_k f_k(Q, R_i)/a_k$ for eight delay-set distances: Chamfer, quantile Wasserstein, gap Wasserstein, log-cardinality ratio, partial assignment, local pattern, MCA complement, and moment difference.

For multimodal matching,

$$c_{ab} = |r_a - r_b| + \lambda_\theta \arccos(\mathbf{u}_a^\top \mathbf{u}_b) + \lambda_P |P_a - P_b|/10, \quad (10)$$

$$z_{ab} = c_{ab}/(1 + c_{ab}), \quad S(z) = (1 - z)^2(1 + 2z). \quad (11)$$

The Beta-shaped row symmetrically averages nearest pair costs weighted by $S(z)$ and adds a normalized cardinality mismatch. “Beta-shaped” is the correct name: the implemented polynomial is not an exact Beta(2, 5) survival integral.

The coherent row uses the receiver’s active-tap array projection distance D_i^{ADP} plus $\lambda_d D_i^{\text{Ch}}$. Survival-CIR combines $D_i^{\text{surv}} = aD_i^{\text{Beta}} + bD_i^{\text{ADP+ToA}} + cD_i^{\text{Ch}}$; residual graph diffuses this score over the RP graph.

PointNet/Deep-Sets encode tokens using

$$h_\ell = \phi(z_\ell), \quad e(Z) = [\text{mean}_\ell h_\ell, \max_\ell h_\ell, \log(1 + |Z|)]. \quad (12)$$

CAEZ-style classification uses $p_i = \text{softmax}(g)_i$ and $\hat{\mathbf{x}} = \sum_i p_i \mathbf{x}_i$. A residual head uses $\hat{\mathbf{x}} = \text{clip}\{\mathbf{a} + \rho \tanh f_\theta(e, \mathbf{a}, d)\}$. RRLE predicts each frozen expert’s error $\hat{e}_k$ and mixes its four coordinates with $\pi_k \propto e^{-\hat{e}_k/T}$. Genuine cross-attention uses

$$\text{Attn}(Q, K, V) = \text{softmax}\left(QK^\top / \sqrt{d} + M\right) V, \quad (13)$$

with padding mask M and no path-order positional encoding. The self-attention candidate model applies the same masked operation within each set, pools, then scores candidates equivariantly. Shuffle, padding, and candidate-permutation tests passed.

2.3 Recipe index for all 25 rows

ID	Artifact key	Beginner-reproducible recipe
M1	majdi_mpurge_map_corrected_prose	Compute corrected s_i above; inverse-weight the best three RPs.
M2	majdi_mca_eps1.6_k1	For each query delay, find its nearest RP delay; sum $(1.6 - d)^2$ only when $d < 1.6$ m; return the highest-scoring RP.
M3	majdi_original_mpurge_P5_corrected	Repeat M1 with total $P = 5$ (half-window 2), $\alpha = .5$, unnormalized pattern, no MAP coverage penalty.
M4	ablation_A_chamfer_inverse_top3	Replace only M1’s score with exact symmetric Chamfer; retain inverse top-three and the common fallback.
M5	ablation_B_temperature_softmax_only	Retain M1 score; replace only inverse weights by development-selected temperature weights $w_i^{(T)}$.
M6	ablation_C_path_count_empty_only	Retain M1 and multiply finite scores by $1 + \eta Q - R_i / \max(Q , R_i)$; give a finite disclosed cost to empty references.
M7	ablation_D_chamfer_temperature_pathcount	Combine M4’s Chamfer, M5’s softmax, and M6’s count/empty rule.
M8	subset_consensus_chamfer	Localize the full set and delete-one subsets (or all size-three subsets for at most six paths); keep hypotheses near their coordinate-wise median and average.
M9	graph_diffusion_corrected	Convert M1 scores to a temperature posterior, run L steps of six-neighbour diffusion G , and take its coordinate mean.
M10	frozen_evomdp_rank	Scale and mix the eight frozen delay distances f_k , temperature-decode the best frozen K RPs; do not refit the genome.
M11	beta_survival_multimodal	Form range/power/AoA pair costs c_{ab} , apply $S(z)$, symmetrize nearest supported costs, add path-count mismatch, inverse-decode three RPs.
M12	coherent_adp_toa	Add coherent active-tap CIR/array distance to weighted delay Chamfer; inverse-decode the best three.
M13	survival_cir_toa	Development-select the nonnegative mixture of M11, M12, and Chamfer, then inverse-decode.
M14	assigned_vt_consensus_aoa	Cluster survey endpoints $\mathbf{v} = \mathbf{x}_i + r\mathbf{u}$ without IDs; query proposals are $\mathbf{v} - r\mathbf{u}$; return the strongest multi-path DBSCAN consensus.
M15	residual_graph_multimodal	Apply M9’s graph diffusion to M13’s multimodal score instead of M1.
M16	delay_feature_extratrees	ExtraTrees on 17 delay quantiles, 7 gap quantiles, count, mean, standard deviation, minimum, and maximum; fit only stored calibration variants.
M17	multimodal_feature_extratrees	Add 7 power quantiles, weighted AoA mean/spread, and 7 normalized CIR summaries to M16; refit the tree ensemble.
M18	rrle_moe	Obtain M1, M4, M10, and M9 coordinates; predict their errors from delay/diagnostic/disagreement features; softmax-mix the experts.
M19	pointnet_delay_direct	Encode unordered scalar-delay tokens by shared MLP plus masked mean/max; directly regress normalized (x, y) .
M20	pointnet_multimodal_direct	Use M19 with token $[r, P, u_x, u_y, u_z]$.
M21	caez_grid_probability_delay	Encode the delay set, output 80 RP logits, train by cross-entropy, and return the posterior coordinate mean.
M22	residual_to_analytic_delay	Feed delay-set encoding, M1 pose, and M1 diagnostics to a bounded residual head.
M23	pointnet_analytic_reranker_delay	Shortlist with M1; separately PointNet-encode query/reference delay sets plus coverage/score/RP coordinate; softmax-average candidates.
M24	cross_attention_multimodal	Let each query path attend to each candidate’s reference paths using noisy range/power/AoA; masked-pool and score candidates.
M25	candidate_self_attention_multimodal	Self-attend within query/reference sets, pool invariantly, then apply a permutation-equivariant candidate attention/reranking head.

2.4 Complete pooled leaderboard

Rank	Method key	Mean	RMSE	Median	P90
1	beta_survival_multimodal	0.2528	0.3731	0.1627	0.5544
2	survival_cir_toa	0.2601	0.3789	0.1716	0.5649
3	residual_graph_multimodal	0.2745	0.4061	0.1940	0.5036
4	candidate_self_attention_multimodal	0.3349	0.4837	0.2156	0.6983
5	cross_attention_multimodal	0.3434	0.4868	0.2406	0.6836
6	assigned_vt_consensus_aoa	0.3463	0.8276	0.0238	1.3415
7	pointnet_multimodal_direct	0.4373	0.5317	0.3697	0.8041
8	multimodal_feature_extratrees	0.4480	0.5359	0.3866	0.8158
9	coherent_adp_toa	0.7746	0.9484	0.6739	1.5223
10	rrle_moe	1.0445	1.2987	0.8504	2.1244
11	ablation_C_path_count_empty_only	1.0589	1.3381	0.8213	2.2701
12	majdi_mpurge_map_corrected_prose	1.0638	1.3429	0.8075	2.2723
13	ablation_B_temperature_softmax_only	1.0721	1.3488	0.8269	2.2389
14	subset_consensus_chamfer	1.0752	1.3369	0.8409	2.2020
15	pointnet_analytic_reranker_delay	1.0961	1.2809	0.9760	2.0162
16	ablation_A_chamfer_inverse_top3	1.1077	1.3979	0.8655	2.3055
17	graph_diffusion_corrected	1.1084	1.3510	0.9460	2.1038
18	residual_to_analytic_delay	1.1252	1.3333	0.9803	2.0701
19	ablation_D_chamfer_temperature_pathcount	1.1345	1.4118	0.9387	2.2924
20	frozen_evomdp_rank	1.1353	1.4131	0.9120	2.2993
21	delay_feature_extratrees	1.2039	1.4055	1.0774	2.2369
22	majdi_mca_eps1.6_k1	1.3143	1.8017	0.8718	3.1994
23	majdi_original_mpurge_P5_corrected	1.3160	1.5544	1.2030	2.4810
24	caez_grid_probability_delay	1.3387	1.5261	1.1798	2.3893
25	pointnet_delay_direct	1.3509	1.5156	1.2350	2.2620

All values are metres over 1,600 identical condition/query blocks. The best three pooled methods are feature-rich; Beta-survival reduces mean error versus corrected Majid by 0.8110 m, query-block bootstrap 95% CI [0.7224, 0.9033] m. Self-attention reduces it by 0.7289 m, independently audited CI [0.6479, 0.8180] m; cross-attention reduces it by 0.7204 m, CI [0.6415, 0.8052] m. Assigned-VT’s 2.38 cm median but 1.34 m P90 reveals rare association failures.

Pure components do not establish a delay-only analytic win: A (Chamfer only) has mean 1.1077 m, reduction -0.0440 m and CI $[-0.1088, 0.0201]$ m; B (softmax only) 1.0721 m, reduction -0.0084 m and CI $[-0.0149, -0.0016]$ m; C (count/empty only) 1.0589 m, reduction 0.0049 m and CI $[-0.0036, 0.0133]$ m; D (all three) 1.1345 m, reduction -0.0708 m and CI $[-0.1373, -0.0044]$ m. Multimodal PointNet beats delay PointNet by 0.9136 m (CI [0.7931, 1.0371] m), and multimodal ExtraTrees beats delay ExtraTrees by 0.7559 m (CI [0.6433, 0.8742] m). The large gain is therefore mainly sensor-enabled; the attention rows show viability, not architecture-only causality.

3 Protocol II: original MPURge VT construction

3.1 Corrected task and evidence boundary

This protocol estimates an *unordered set of VT coordinates*, not a receiver coordinate. Transmitter identity is observable because elementary transmitters use separable waveform channels; it is retained throughout matching, grouping, support counting, and evaluation. True VT coordinates, cardinality, ray identity, and query truth are never primary-method inputs. The paper’s total window $P = 5$ is implemented as half-width $p = 2$. Current conflicts are resolved in increasing composite dissimilarity, and both the literal printed crossing test and geometric order-reversal test are recorded. Both overlapping-star and conflict-resolved connected-component grouping are bracketed. Beta support is swept until the retained set is empty.

The intended full design contains 24 independent scene/noise blocks: six scene seeds times $\sigma_r \in \{0, 0.25, 1, 3\}$ m, 36 survey positions, 20 held-out queries, 90 differentiable-assignment steps with two restarts, 120 epochs for each self-supervised set model, and a 31×31 downstream localization grid. The process stopped after checkpoint stages for 7/24 blocks and wrote no block metrics, aggregate, report, or manifest. Those checkpoints are not a partial leaderboard.

The completed diagnostic smoke used one scene, $\sigma_r \in \{0, 0.25\}$ m, 20 survey positions, six queries, eight neural epochs, 12 assignment steps, and a 15×15 grid. Its one-scene intervals are necessarily degenerate.

3.2 Construction and scoring equations

For nearby survey anchors $\mathbf{x}_i$, match unordered indirect ranges with the corrected MPURge dissimilarity from Protocol I (or the PBA L2 size- unification control). A conflict-compatible group $\mathcal{G}$ is converted to a VT by robust range

trilateration,

$$\hat{\mathbf{v}}_G = \arg \min_{\mathbf{v} \in \mathbb{R}^2} \sum_{(i,a) \in G} (\|\mathbf{v} - \mathbf{x}_i\|_2 - r_{ia})^2. \quad (14)$$

For a candidate $\mathbf{v}$, support is the number of distinct anchors whose measured range agrees within the frozen gate. With mean support $\bar{s}$, the descriptive sweep retains $\{\mathbf{v} : s(\mathbf{v}) \geq \beta \bar{s}\}$ and increases β until this set is empty.

Multimodal inverse proposals are

$$\mathbf{v}_{ia} = \mathbf{x}_i + r_{ia} \mathbf{u}_{ia}. \quad (15)$$

Complete-link clusters with at least three anchors are power-weighted. The survival version keeps a cluster only when $\Pr[p > 0.15 \mid m, n] = 1 - F_{\text{Beta}(1+m, 1+n-m)}(0.15) \geq 0.90$. The Bernoulli/RFS tracker updates a matched track existence by $q \leftarrow q + 0.35(1 - q)$ and a missed track by $q \leftarrow 0.94q$; it emits tracks with $q \geq 0.55$ and support from at least three anchors.

Unknown-cardinality differentiable assignment tests candidate counts, uses a soft symmetric set loss

$$L_K = \frac{1}{N} \sum_i \left[\frac{1}{|R_i|} \sum_{r \in R_i} \text{smin}_{k,\tau} |r - d_{ik}| + \frac{1}{K} \sum_k \text{smin}_{r \in R_i, \tau} |d_{ik} - r| \right] + \text{BIC}(K), \quad (16)$$

where $d_{ik} = \|\mathbf{x}_i - \mathbf{v}_k\|_2$ and $\text{smin}_\tau(a) = -\tau \log \sum_j e^{-a_j/\tau}$. DeepSets and self-attention instead predict a variable-existence set and minimize only range/AoA reconstruction on the allowed survey; no VT labels are used.

Evaluation first maximizes gated bipartite match cardinality and only then minimizes matched distance. If TP, FP, and FN are the resulting counts,

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2TP}{2TP + FP + FN}. \quad (17)$$

Symmetric Chamfer, Hausdorff, and downstream grid-localization error prevent a high F1 from hiding poor coordinates or an unusable VT set.

3.3 Recipe index for every diagnostic row

ID	Artifact key	Reproduction recipe
V1	<code>bernoulli_rfs_multimodal</code>	Process survey anchors sequentially; form range/AoA proposals, gate to existing tracks, update coordinate/existence, decay missed tracks, retain $q \geq .55$ with three-anchor support.
V2	<code>beta_marginal_survival_multimodal</code>	Cluster range/AoA proposals as above; integrate the observed support under the Beta posterior and retain survival probability at least .90.
V3	<code>aoa_power_inverse_consensus</code>	Complete-link cluster $\mathbf{x}_i + r\mathbf{u}$ proposals and power-weight each cluster; require at least three distinct anchors.
V4	<code>corrected_pba_order_components</code>	L2 size unification, geometric crossing, connected-component groups, trilateration, dynamic beta.
V5	<code>corrected_mpurge_order_components</code>	L1 MPURge size unification, geometric crossing, connected components, trilateration, dynamic beta.
V6	<code>cycle_consistent_delay</code>	Repeatedly sample three anchors/paths, trilaterate, require range support on other anchors, complete-link cluster accepted hypotheses.
V7	<code>corrected_pba_printed_components</code>	V4 with literal printed crossing instead of geometric order crossing.
V8	<code>corrected_mpurge_printed_components</code>	V5 with literal printed crossing.
V9	<code>corrected_mpurge_order_star</code>	V5 with overlapping one-hop star grouping instead of connected components.
V10	<code>corrected_pba_order_star</code>	V4 with overlapping star grouping.
V11	<code>corrected_pba_printed_star</code>	PBA, printed crossing, star grouping.
V12	<code>corrected_mpurge_printed_star</code>	MPURge, printed crossing, star grouping.
V13	<code>diffassign_bic_multimodal</code>	Optimize L_K over candidate K , include noisy AoA reconstruction, choose K by survey-only BIC.
V14	<code>selfsup_deepsets_multimodal</code>	Shared token MLP, masked mean/max pooling, variable-existence VT decoder; fit survey range/AoA reconstruction only.
V15	<code>selfsup_attention_multimodal</code>	Replace V14's token interaction by masked self-attention without positional encoding; keep the same self-supervised loss.
V16	<code>diffassign_bic_delay</code>	V13 using surveyed delays and coordinates only.

3.4 Complete smoke leaderboard: diagnostic, not final

Rank	Method	F_1	Precision	Recall	Chamfer	Hausdorff	Loc. mean
1	bernoulli_rfs_multimodal	.9554	.9150	1.0000	.0559	.1986	.2864
2	beta_marginal_survival_multimodal	.8515	.7521	1.0000	.0807	.1882	.2853
3	aoa_power_inverse_consensus	.7729	.6390	1.0000	.0948	.2874	.3044
4	corrected_pba_order_components	.2572	.2910	.3125	8.7439	32.8345	5.1719
5	corrected_mpurge_order_components	.2556	.2897	.3125	8.6965	32.8345	5.0178
6	cycle_consistent_delay	.1584	.0905	.6875	3.2785	21.4026	.9159
7	corrected_pba_printed_components	.1442	.0874	.4375	4.3695	24.2796	.6892
8	corrected_mpurge_printed_components	.1412	.0851	.4375	4.2370	24.2796	.6799
9	corrected_mpurge_order_star	.0867	.0455	.9375	2.3526	17.8574	.3959
10	corrected_pba_order_star	.0860	.0451	.9375	2.3336	17.7446	.3724
11	corrected_pba_printed_star	.0758	.0395	.9375	2.6123	22.0860	.7048
12	corrected_mpurge_printed_star	.0756	.0394	.9375	2.6459	22.7884	.7124
13	diffassign_bic_multimodal	.0556	.0500	.0625	8.8376	21.4899	2.1195
14	selfsup_deepsets_multimodal	.0000	.0000	.0000	13.9164	22.7315	8.1569
15	selfsup_attention_multimodal	.0000	.0000	.0000	14.8455	30.7247	10.9501
16	diffassign_bic_delay	.0000	.0000	.0000	10.7496	21.8988	3.5806

Distances are metres. These numbers support triage only: the analytic multimodal RFS/survival/consensus branches deserve full comparison, while the eight-epoch neural failure is inconclusive and cannot justify rejecting set attention or DeepSets.

PENDING FULL-RUN INPUT HOOK. Place the final, audited 24-block table in `research/four_paper_report/mpurge_vt_full_leaderboard.tex`. It must state run status, scene/noise block count, source/receiver hashes, raw-row paths, primary versus oracle rows, and scene-block bootstrap intervals.

4 Protocol III: SMART-LEA

4.1 Frozen full reconstruction

The paper describes an irregular two-dimensional floorplan with four physical transmitters and four first-order VTs per transmitter (20 anchors total). The replacement uses a disclosed 25×25 m rectangle, invented Tx coordinates, and four wall images per Tx. A single stored 100×100 survey at 0.25 m provides strict nested maps of 10,000, 2,500, 289, and 100 RPs for spacings 0.25, 0.5, 1.5, 2.5 m. No coarse method sees dense-map rows. Each of seeds 29, 47, and 71 has 1,000 fixed off-grid queries in clean and matched-noisy conditions. Matched noise means 20 dB receiver SNR plus 0.25 m range, 2 dB power, and 5° direction perturbations after extraction. Learned selection holds out complete 5 m spatial blocks, then a fresh model is fitted at the permitted spacing.

SMART’s literal primary input is an initial MCA coordinate, the coordinates of the 20 paper-known real/virtual transmitters, and the measured delay set. Richer map challengers instead receive only their declared same-acquisition features and RP coordinates. The separately labelled legacy four-solve row uses the observable physical-Tx channel partition; it is not Algorithm 1 and is excluded from primary winner claims.

4.2 Core equations and method recipes

At iteration t , SMART recalculates every anchor range and jointly matches all 20 values:

$$\tilde{r}_j^{(t)} = \|\tilde{\mathbf{x}}^{(t)} - \mathbf{v}_j\|_2, \quad \tilde{\mathbf{x}}^{(t+1)} = \arg \min_{\mathbf{x}} \sum_{(a,j) \in \mathcal{M}^{(t)}} (\|\mathbf{x} - \mathbf{v}_j\|_2 - q_a)^2. \quad (18)$$

The fixed-height correction evaluates $\sqrt{(x - v_{jx})^2 + (y - v_{jy})^2 + (z_r - v_{jz})^2}$ in both matching and least squares. Total $P = 5$ means half-window $p = 2$. The printed crossing test $(q_a - r_b)(q_c - r_d) < 0$ and geometric test $(q_a - q_c)(r_b - r_d) < 0$ are separate rows.

The additional analytic scores reuse MCA, Chamfer, graph diffusion, subset consensus, and EvoMDP from Protocol I. The combined delay score is

$$D_i = \beta \tilde{D}_i^{\text{Ch}} + (1 - \beta) \left[\frac{1}{2} \tilde{D}_i^{W_1} + \frac{1}{2} \left(1 - S_i^{\text{MCA}} / \max_j S_j^{\text{MCA}} \right) \right]. \quad (19)$$

For unit-normalized array-PDP $\mathbf{c}$, coherent distance is $D_i^{\text{CIR}} = 1 - \mathbf{c}_q^\top \mathbf{c}_i$; delay-CIR uses $\eta \tilde{D}_i^{\text{Ch}} + (1 - \eta) \tilde{D}_i^{\text{CIR}}$. Multimodal assignment uses Hungarian matching with

$$C_{ab} = |r_a - r_b|/(0.5 \text{ m}) + |P_a - P_b|/(3 \text{ dB}) + \arccos(\mathbf{u}_a^\top \mathbf{u}_b)/(10^\circ), \quad (20)$$

plus an unmatched-path penalty.

Survey-built VTs minimize a robust range residual over persistent tracks,

$$\hat{\mathbf{v}}_k = \arg \min_{v_x, v_y, h \geq 0} \sum_{i \in \mathcal{T}_k} \rho \left(\sqrt{(x_i - v_x)^2 + (y_i - v_y)^2 + h^2} - r_{ik} \right). \quad (21)$$

The differentiable variant alternately Sinkhorn-normalizes $A_{iak} \propto e^{-|r_{ia} - d_{ik}|/\tau}$ and minimizes the robust assignment-weighted range loss. True VTs are used only after fitting for an audit error.

ID	Artifact key	Reproduction recipe
S1	majid_mca_eps1	MCA with $\epsilon = 1 \text{ m}$, return the best RP; room centre on no evidence.
S2	majid_smart_lea1_joint_totalP5_printed	MCA start, one joint 20-anchor SMART iteration, total $P = 5$, printed crossing.
S3	majid_smart_lea4_joint_totalP5_printed	Repeat S2 for four iterations.
S4	smart_lea4_joint_totalP5_geometric_cross	S3 with geometric/order crossing only.
S5	smart_lea4_joint_cited_halfwindow5_printed	S3 with half-window 5 (total 11) to bracket the cited-parameter ambiguity.
S6	smart_lea4_fixed_height_totalP5_geometric	S4 with the known receiver height in three-dimensional anchor ranges.
S7	legacy_per_tx_fusion_totalP5_printed	Split by observable physical Tx, run four five-anchor solves, robustly fuse; diagnostic only.
S8	corrected_mpurge_map_prefilter24	Exact Chamfer shortlist 24 RPs, evaluate corrected coverage-penalized MPUrges only there, inverse top-three.
S9	symmetric_chamfer_inn	Compute symmetric delay Chamfer and return its single best RP.
S10	chamfer_temperature	Development-select $K \in \{3, 5, 9\}$ and T , then softmax-average the top Chamfer RPs.
S11	combined_delay_score	Median-normalize Chamfer, Wasserstein, and MCA complement; select β on spatial development blocks.
S12	subset_consensus	Full/delete-one delay hypotheses, robust inlier filter, mean coordinate.
S13	graph_diffusion	Convert Chamfer scores to a grid posterior and diffuse through a normalized 3×3 kernel.
S14	evomdp_rank_frozen_transfer	Apply the frozen eight-distance genome without refitting its weights.
S15	coherent_array_pdp_temperature	Cosine distance between noisy normalized array-PDP vectors, temperature-decode the map.
S16	range_power_aoa_assignment	Hungarian range/power/AoA matching with unmatched penalty, temperature-average best RPs.
S17	delay_cir_combined	Development-select the normalized Chamfer/PDP-cosine mixture η .
S18	pointnet_delay	Shared delay encoder, masked mean/max/count pooling, direct coordinate head.
S19	pointnet_range_power_aoa	S18 with noisy $[r, P, \mathbf{u}]$ tokens.
S20	query_reference_attention_delay	Embed query and shortlisted reference delay sets; query-to-reference multihead attention; candidate softmax.
S21	query_reference_attention_multimodal	S20 with noisy range/power/AoA tokens.
S22	candidate_pointnet_reranker_delay	Independently PointNet-encode query/candidate sets and score encoding difference, RP coordinate, and Chamfer diagnostic.
S23	candidate_path_cross_attention_multimodal	Each query path cross-attends to each candidate's paths; masked-pool and softmax-average coordinates.
S24	caez_style_delay_probability_mlp	Fixed delay vector to one logit per allowed RP; posterior coordinate mean.
S25	caez_style_cir_probability_mlp	S24 with a 256-element pooled noisy array-PDP vector.
S26	extra_trees_delay_features	ExtraTrees on 20 padded delays, 19 gaps, count, mean, and standard deviation.
S27	analytic_anchor_residual	Chamfer-temperature anchor plus delay features/score margin/local spread; bounded 5 m tanh correction.
S28	rrle_moe	Route among Chamfer, ExtraTrees, delay PointNet, and analytic residual using spatially held-out predicted expert errors.
S29	lea4_mpurge_track_vts	Build survey-neighbour associations with corrected MPUrges, robustly fit persistent VTs, feed them to fixed-height LEA4.
S30	lea4_diffassign_vts	Initialize S29 VTs, refine with Sinkhorn range assignment on CUDA, feed the fitted set to the same LEA4.

4.3 Complete SMART macro leaderboard

The table macro-averages mean error across the four spacings, separately for clean and matched-noisy cells; the final column equally averages all eight condition/spacing cells. Every cell contains 3,000 protected query rows (three seeds times 1,000 queries). S7 is shown for completeness but marked diagnostic.

Rank	Method	Track	Clean	Noisy	All
1	range_power_aoa_assignment	multimodal	.3304	.5219	.4261
2	delay_cir_combined	multimodal	1.7888	1.7246	1.7567
3	query_reference_attention_multimodal	learned multimodal	1.4695	2.3953	1.9324
4	coherent_array_pdp_temperature	multimodal	2.1407	2.0927	2.1167
5	symmetric_chamfer_1nn	primary	2.4668	2.7267	2.5967
6	corrected_mpurge_map_prefilter24	primary	2.6957	3.0294	2.8626
7	pointnet_range_power_aoa	learned multimodal	2.7296	3.1246	2.9271
8 [†]	legacy_per_tx_fusion_totalP5_printed	diagnostic	2.8405	3.4094	3.1249
9	subset_consensus	primary	2.9530	3.3626	3.1578
10	evomdp_rank_frozen_transfer	frozen transfer	3.0084	3.4497	3.2290
11	combined_delay_score	primary	2.8842	4.6757	3.7799
12	chamfer_temperature	primary	3.6735	4.1436	3.9086
13	candidate_pointnet_reranker_delay	learned	3.7571	4.3277	4.0424
14	candidate_path_cross_attention_multimodal	learned multimodal	4.0339	4.1794	4.1067
15	query_reference_attention_delay	learned	3.9112	4.3843	4.1477
16	smart_lea4_fixed_height_totalP5_geometric	analytic correction	3.8254	4.6683	4.2469
17	smart_lea4_joint_totalP5_geometric_cross	ambiguity bracket	3.8375	4.6767	4.2571
18	analytic_anchor_residual	learned hybrid	3.7423	5.0162	4.3792
19	majid_smart_lea1_joint_totalP5_printed	paper primary	3.9455	4.8167	4.3811
20	majid_smart_lea4_joint_totalP5_printed	paper primary	3.9393	4.8266	4.3829
21	smart_lea4_joint_cited_halfwindow5_printed	ambiguity bracket	4.0092	4.8396	4.4244
22	majid_mca_eps1	paper primary	4.0484	4.8740	4.4612
23	graph_diffusion	primary	4.3962	5.2339	4.8151
24	caez_style_cir_probability_mlp	learned multimodal	5.1453	5.1844	5.1648
25	extra_trees_delay_features	learned	3.6889	7.4714	5.5802
26	lea4_mpurge_track_vts	survey geometry	7.8734	7.8749	7.8742
27	lea4_diffassign_vts	survey geometry	7.6689	8.2406	7.9547
28	rrle_moe	learned hybrid	6.2895	9.9100	8.0998
29	caez_style_delay_probability_mlp	learned	7.5507	9.4283	8.4895
30	pointnet_delay	learned	9.2694	9.3757	9.3225

[†]Not eligible for the primary winner because it changes the paper’s joint 20-anchor operation into four Tx-partitioned solves. Metres throughout.

The cell-wise winner changes with density and noise: fixed-height geometric SMART wins clean 0.25 m (0.0198 m) and 0.5 m (0.0562 m); range/power/AoA assignment wins clean 1.5 m (0.3197 m), clean 2.5 m (0.8576 m), noisy 1.5 m (0.4931 m), and noisy 2.5 m (1.0460 m); delay-CIR wins noisy 0.25 m (0.1338 m); multimodal query-reference attention wins noisy 0.5 m (0.2266 m). This is why a single macro rank must not replace per-density reporting.

4.4 All 30 methods in all eight SMART cells

Method	C-.25	C-.5	C-1.5	C-2.5	N-.25	N-.5	N-1.5	N-2.5
range_power_aoa_assignment	.060	.084	.320	.858	.241	.308	.493	1.046
delay_cir_combined	.088	.207	2.607	4.254	.134	.229	2.506	4.030
query_reference_attention_multimodal	.069	.129	.570	5.110	.181	.227	2.032	7.141
coherent_array_pdp_temperature	.164	.483	3.472	4.444	.165	.599	3.383	4.224
symmetric_chamfer_lnn	.102	.202	2.720	6.843	.320	.499	3.204	6.883
corrected_mpurge_map_prefilter24	.041	.111	3.820	6.810	.215	.404	4.443	7.056
pointnet_range_power_aoa	.661	1.109	6.830	2.318	1.373	1.844	5.409	3.872
legacy_per_tx_fusion_totalP5_printed	.071	.142	4.575	6.574	.776	1.413	4.852	6.597
subset_consensus	.044	.190	4.488	7.090	.282	.797	5.125	7.245
evomdp_rank_frozen_transfer	.048	.192	4.411	7.382	.318	.843	5.093	7.545
combined_delay_score	.145	.439	3.468	7.485	.346	1.883	7.416	9.058
chamfer_temperature	.060	.164	5.999	8.471	.303	.804	7.281	8.186
candidate_pointnet_reranker_delay	.059	.127	6.081	8.762	.337	1.045	7.018	8.911
candidate_path_cross_attention_multimodal	.064	.167	7.031	8.874	.213	.455	7.008	9.041
query_reference_attention_delay	.065	.191	6.682	8.707	.342	1.137	7.149	8.910
smart_lea4_fixed_height_totalP5_geometric	.020	.056	6.080	9.146	.845	1.787	6.687	9.355
smart_lea4_joint_totalP5_geometric_cross	.031	.068	6.089	9.163	.851	1.789	6.702	9.365
analytic_anchor_residual	.085	.223	6.117	8.544	1.510	2.824	7.455	8.275
majid_smart_lea1_joint_totalP5_printed	.048	.107	6.216	9.411	.911	1.862	6.838	9.656
majid_smart_lea4_joint_totalP5_printed	.037	.079	6.207	9.434	.922	1.843	6.812	9.729
smart_lea4_joint_cited_halfwindow5_printed	.037	.074	6.360	9.566	.889	1.852	6.930	9.688
majid_mca_eps1	.102	.238	6.385	9.469	.993	1.967	6.932	9.604
graph_diffusion	.060	.166	7.936	9.423	1.157	1.362	9.185	9.233
caez_style_cir_probability_mlp	.446	1.327	9.374	9.434	.501	1.510	9.336	9.391
extra_trees_delay_features	.618	1.853	5.042	7.243	5.625	6.840	8.412	9.009
lea4_mpurge_track_vts	4.095	4.910	9.004	13.484	4.723	5.780	9.763	11.233
lea4_diffassign_vts	3.904	4.104	9.839	12.829	3.929	5.739	10.414	12.879
rrle_moe	.265	1.456	11.000	12.438	5.666	8.531	12.875	12.568
caez_style_delay_probability_mlp	3.529	7.773	9.403	9.498	9.558	9.249	9.401	9.505
pointnet_delay	8.935	9.314	9.412	9.416	9.285	9.312	9.419	9.487

C denotes clean, N matched-noisy, and the suffix is map spacing in metres. Entries are mean position error in metres, read directly from the 240 summary rows in `smart_fair_full/results.json`.

Condition	Spacing	Winner	Mean	RMSE	Median	P90
clean	.25	fixed-height geometric LEA4	.0198	.0224	.0187	.0341
clean	.5	fixed-height geometric LEA4	.0562	.8452	.0187	.0342
clean	1.5	range/power/AoA assignment	.3197	.3715	.2813	.5385
clean	2.5	range/power/AoA assignment	.8576	1.2751	.6847	1.3894
noisy	.25	delay-CIR combined	.1338	.1524	.1240	.2276
noisy	.5	multimodal query-reference attention	.2266	.3131	.2038	.3850
noisy	1.5	range/power/AoA assignment	.4931	.5753	.4482	.8863
noisy	2.5	range/power/AoA assignment	1.0460	1.9119	.7351	1.5894

Table 9: Per-cell winner distributions. The clean 0.5 m LEA4 row has an excellent median/P90 but RMSE 0.8452 m, exposing a rare severe tail.

5 Protocol IV: two-room CNN fingerprinting

5.1 Printed protocol and corrected reconstruction

The paper uses Room A (20×8 m) and Room B (16×12 m), 20 RPs, 60 AP positions, and seven obstacle configurations. This grants $20 \times 60 \times 7 = 8,400$ labelled calibration acquisitions per room. It tests four obstacle-count conditions, two APs per condition, and 400 uniform points per condition: 1,600 locked off-grid queries per room. The printed input contains the nine ranges selected by *observed* power, all 20 RP coordinate triplets, and the AP coordinate, reshaped to 6×12 .

The replacement preserves those counts and room sizes, but its disclosed AP positions, obstacle planes, and image-source propagation are not NIST Q-D. Each legal challenger receives no more than the printed 8,400 acquisitions; it may use noisy delay, power, receive AoA, and array CIR extracted there. It may not add positions, simulations, clean channels, ray/VT IDs, query truth, or hidden obstacle identity in the deployable bracket.

Within each AP/configuration group, 16 of 20 RP acquisitions fit and four validate. Candidate/residual/RRLE pseudo-queries compare only with fit-map rows and explicitly exclude the source acquisition:

$$j \notin \mathcal{C}(Q_j), \quad \mathcal{C}(Q_j) \subseteq \mathcal{I}_{\text{fit}}. \quad (22)$$

Locked queries may use the complete acquired map only after all choices freeze.

Reference methods have two non-equivalent information brackets. The paper-matched sensitivity bracket supplies the true obstacle condition $R_i^{\text{oracle}} = R_{aic_q}$ and is explicitly privileged. The deployable environment-blind bracket selects, per RP,

$$c_i^* = \arg \min_{c \in \{0, \dots, 6\}} D(Q, R_{aic}), \quad R_i^{\text{blind}} = R_{aic_i^*}, \quad (23)$$

without exposing the label. Direct models use neither bracket and are emitted once as protocol-independent rows.

5.2 Equations and the 25-method recipe index

Fit-only min-max scaling converts observed-power-ranked delays to $\tilde{d}_{1:9}$. The paper vector

$$v = [\tilde{d}_{1:9}, \tilde{\mathbf{x}}_{1:20}^{\text{RP}}, \tilde{\mathbf{a}}] \in \mathbb{R}^{72} \quad (24)$$

is reshaped column-major to $1 \times 6 \times 12$. The classifier outputs $p_i = \text{softmax}(g)_i$ and $\hat{\mathbf{x}} = \sum_i p_i \mathbf{x}_i$ with cross-entropy; the regressor minimizes the printed batch coordinate RMSE. Three 2×1 convolution blocks yield 36,148 classifier and 36,190 regressor parameters; the paper’s reported regressor count differs by 21 and remains unresolved.

Map methods reuse MCA, corrected MPUre, Chamfer, subset consensus, graph diffusion, and EvoMDP from Protocol I. Multimodal pair costs are

$$C_{ab}^{rP} = |r_a - r_b| + |P_a - P_b|/8, \text{quad} C_{ab}^{r\theta} = |r_a - r_b| + 1.5 \arccos(\mathbf{u}_a^\top \mathbf{u}_b). \quad (25)$$

For active CIR taps, coherent ADP uses

$$\rho_n = \frac{|\mathbf{q}_n^H \mathbf{r}_n|^2}{\|\mathbf{q}_n\|_2^2 \|\mathbf{r}_n\|_2^2}, \quad D_{\text{ADP}} = 1 - \frac{\sum_n \sqrt{E_{qn} E_{rn}} \rho_n}{\sum_n \sqrt{E_{qn} E_{rn}}}. \quad (26)$$

Survey-built inverse consensus clusters $\mathbf{v} = \mathbf{x}_i + r\mathbf{u}$ and proposes query coordinates $\hat{\mathbf{x}}_{\ell k} = \mathbf{v}_k - r_{\ell} \mathbf{u}_{\ell}$. Beta-marginal visibility contributes $\log p(m | n) = \log B(2 + m, 1 + n - m) - \log B(2, 1)$.

ID	Planned artifact key	Reproduction recipe
C1	majid_cnn_classifier	Build the printed 72-vector, column-major reshape, three conv blocks, 20-way cross-entropy, posterior RP mean.
C2	majid_cnn_regressor	Same input/backbone as C1, two-coordinate head, printed batch-RMSE loss.
C3–4	majid_mca_eps1, majid_mca_eps2	MCA at $\epsilon = 1$ and 2 m under the declared map bracket.
C5	corrected_mpurge_map	Coverage-corrected MPURge, $p = 6$, normalized pattern, printed crossing, inverse top-three.
C6	symmetric_chamfer_inverse3	Symmetric delay Chamfer, inverse-weight the best three.
C7	symmetric_chamfer_softmax3	C6 with fixed $T = .03$ softmax over the best three.
C8	subset_consensus	Full and path-subset Chamfer hypotheses, median-radius inlier fusion.
C9	graph_diffusion	Rank-transform MPURge evidence, diffuse over the two-nearest-RP graph, blend 75% analytic and 25% graph mode.
C10	evomdp_frozen_transfer	Frozen eight-distance mixture; fit only feature scales, then top-three temperature decode.
C11	range_power_chamfer	Symmetric nearest-neighbour matching with C^{rP} ; top-three $T = .15$ head.
C12	range_aoa_chamfer	Same with $C^{r\theta}$.
C13	dichasus_coherent_adp_8nn	Noise-gated coherent ADP, spatially distinct best eight RPs, temperature barycentre.
C14	assigned_vt_inverse_consensus	Survey-build VTs from range/AoA proposals, form query inverse proposals, geometric-median consensus blended with a coarse delay pose.
C15	beta_marginal_vt_survival	One-to-one inverse-proposal/VT assignment plus Beta-integrated support and unmatched penalties, top-three decode.
C16	pointnet_delay	Shared scalar-delay encoder, masked mean/max/count, AP context, direct coordinate head.
C17	pointnet_multimodal	C16 with noisy range/power/AoA tokens.
C18	set_attention_multimodal	Two masked four-head self-attention blocks without positional encoding, invariant pooling, direct coordinate head.
C19	candidate_pointnet_reranker	Independently set-encode query/reference, add coordinate/AP/analytic score, guarded candidate softmax.
C20	candidate_set_attention	C19 with self-attended query/reference encoders.
C21	genuine_candidate_cross_attention	Each query path attends to candidate paths; pool $[Q, A, Q - A]$ with map/AP diagnostics and guarded logits.
C22	caez_cir_probability_mlp	Pool noisy CIR magnitude to 64 bins, concatenate AP coordinate, output 20 RP probabilities and their coordinate mean.
C23	analytic_anchor_residual	Environment-blind MPURge anchor plus multimodal PointNet and score margin; bounded 0.25 tanh update.
C24	survival_cir_toa_extratrees	ExtraTrees on 32 CIR energy-survival samples, first ToA, path count, and AP coordinate.
C25	rrle_map_aware_moe	Freeze environment-blind MPURge, Chamfer, and ADP experts; learn validation-error/disagreement router; softmax-mix coordinates.

PENDING FULL-RUN INPUT HOOK. Place the corrected full table in `research/four_paper_report/cnn_full_leaderboard.tex`. It must report Room A/B separately; oracle-condition, environment-blind, and protocol-independent provenance; all 25 unique methods without duplicated predictions; AP-within-condition paired intervals; seed counts; raw-row, checkpoint, source, receiver, and acquisition hashes.

The superseded `cnn_fair_smoke` predates strict leave-self validation and provenance corrections; none of its values enters this report. Published context only: Room A classifier 1.82 m, regressor 1.84 m, MCA-1 2.58 m; Room B classifier 2.14 m, regressor 1.47 m, MCA-1 3.65 m. These are paper values, not replacement rows.

6 Compact causal and multilevel evidence

This section is intentionally limited to at most two pages and is not inserted into any static-protocol leaderboard. Temporal filtering has a different information budget; active multilevel calibration has a different output question.

6.1 Confirmed positive causal sequential result

G2 wraps the frozen corrected-MPUrge RP score vector. Rank scores form the emission and surveyed RP distances form the transition:

$$\ell_t(i) = \exp\left[-\beta \frac{\text{rank}(s_t(i))}{N-1}\right], \quad (27)$$

$$T_{ij} \propto \exp[-\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 / (2\sigma_T^2)], \quad (28)$$

$$p_t = \text{normalize}\{\ell_t \odot (T^\top p_{t-1})\}, \quad \hat{\mathbf{x}}_t = \sum_{i \in \text{top } K(p_t)} \frac{p_t(i)}{\sum_{j \in K} p_t(j)} \mathbf{x}_i. \quad (29)$$

Earlier frames froze $\beta = 8$, $\sigma_T = .6$ m, zero extra stay weight, and $K = 5$ before three shifted 200-frame confirmation segments were opened. The predictor uses only past belief, current analytic scores, and RP coordinates; no future observation or query coordinate enters it.

Method	Mean	Median	P90
G2 causal analytic HMM/particle bank	.3802	.2793	.7907
All-path PointNet	.8121	.3405	1.3887
Corrected MPUrge single-frame	.9654	.3503	2.4194

Across 1,350 matched rows / 270 physical frames, G2 reduces mean error versus PointNet by 0.4319 m, frame-block CI [0.3541, 0.5137] m, and versus static MPUrge by 0.5852 m, CI [0.4958, 0.6783] m. This is a confirmed positive result, but only for ordered trajectories.

6.2 Honest causal multilevel calibration: useful, not a confirmed win

Nine nested acquisition policies choose 12, 20, or 30 of 72 survey candidates without reading any unselected fingerprint: spatial farthest, Chebyshev, polynomial Leja, Smolyak, Sobol, GP variance, Bayesian quadrature, FEM variation, and surrogate disagreement. Each policy renders the missing map with its development-frozen polynomial/FEM/RBF choice and localizes 36 protected queries. The causal acquisition oracle recorded zero unselected reads, no dense-map selection, no query access, and nested paths.

Budget	Lowest observed policy	Mean	Spatial mean	Reduction	95% CI
12	spatial farthest	1.5251	1.5251	.0000	[0, 0]
20	causal FEM variation	1.4744	1.5462	.0718	[−.1534, .3035]
30	causal GP variance	1.3434	1.4271	.0837	[−.1002, .2674]

The positive point estimates at budgets 20 and 30 have intervals crossing zero, so the correct conclusion is neutral rather than a win. An earlier “goal-surplus” result is retracted as calibration-savings evidence because it read every unselected fingerprint. Recent quadrature/multifidelity ideas are therefore not promoted in the headline report: under the honest oracle they did not show a great confirmed result.

7 Release and audit ledger

Evidence block	Status	Auditable scale	Release ruling
MPUrge-MAP full	PASS with disclosed post-run source mutation	40,000 primary rows; 33,600 learned seed rows; 16 conditions; 25 methods	Publish as replacement-scene evidence. All 34 focused checks and invariance/edge tests pass.
SMART full	FULL_COMPLETE; integrity PASS	720,000 raw rows; 24 scenarios; 30 methods	Publish as replacement-scene evidence; retain legacy Tx-partition row as diagnostic.
Original MPUrge smoke	QUICK_SMOKE_COMPLETE	1 scene; 2 noise levels; 16 primary rows; all 12 stored hashes recomputed	Diagnostic triage only; no final rank claim or neural rejection.
Original MPUrge full	INCOMPLETE	7/24 checkpoint stages; no metrics/results/manifest	Do not publish a partial leaderboard; use the explicit input hook after an atomic resumable rerun.
CNN legacy smoke	SUPERSEDED	Old quick artifact	Do not quote; it predates leave-self and provenance fixes.
CNN corrected full	PENDING	Hook expects two rooms and provenance-separated rows	No reconstruction performance claim until raw rows, audit, and hashes exist.
G2 causal	PASS	1,350 rows / 270 frames	Publish separately as sequential evidence, never in the static table.
Causal multilevel	Integrity PASS; no confirmed gain	27 policy-budget summaries; 36 protected queries each	Publish the honest neutral finding; never restore the invalid oracle win.

7.1 Hash and provenance resolution

The MAP manifest preserves receiver SHA-256 7e9909312d9f9330e5e3d37f7ab3962732b2500503e2c1113e09c3f1d18da8f6; the SMART manifest preserves its imported receiver SHA-256 beginning 5a4867f3d1363c54. A concurrent task changed the shared receiver after those completed runs and restored its semantics without byte-identical text; the current file hashes to 10962bf2ab1edf3495879c2693f3e75ad1aeabf68c5af359d9f108645d3b5e67. The manifests must not be rewritten to imply that current bytes generated old rows. Completed result/checkpoint hashes remain intact; the mismatch is resolved by disclosure, not by altering run outputs.

7.2 Exact evidence paths

All paths below are relative to the project root.

Purpose	Path
MAP aggregates	research/four_paper_report/corrected_mpurge_map_full/aggregate_metrics.json
MAP paired intervals	research/four_paper_report/corrected_mpurge_map_full/paired_bootstrap_vs_corrected_majid.json
MAP raw/seed rows	research/four_paper_report/corrected_mpurge_map_full/raw_predictions.jsonl; learned_seed_predictions.jsonl
MAP audit/provenance	research/four_paper_report/corrected_mpurge_map_full/focused_result_audit.json; provenance_resolution.json
SMART result/raw/audit	research/four_paper_report/smart_fair_full/results.json; raw_predictions.jsonl.gz; integrity_audit.json
VT smoke	research/four_paper_report/corrected_mpurge_vt_smoke/results.json; SHA256SUMS.txt
VT interrupted full	research/four_paper_report/corrected_mpurge_vt_full/config.json; partial checkpoints are non-results
CNN corrected runner	experiments/four_paper_report/run_cnn_fair.py; test_cnn_fair.py
Four-paper freeze/audits	research/four_paper_report/protocol-freeze.md; mpurge_protocol_audit.md; smart_cnn_protocol_audit.md
Method dependency inventory	research/four_paper_report/requested_method_inventory.md
G2 causal evidence	research/campaign_8plus4/g2_replication_results.json; g2_replication_rows.jsonl
Multilevel evidence	research/multilevel_idp/confirmation_results.json; confirmation_rows.jsonl

7.3 Minimal reproducibility checklist

1. Materialize and hash survey/query observations before training. Verify independent survey/query corruption and absence of forbidden identity fields.
2. Enforce the exact paper-specific acquisition count. At sparse SMART spacing, subset the stored dense acquisition before any fit; for CNN, never exceed the printed 8,400 acquisitions per room.
3. Fit normalization and choose all hyperparameters on grouped calibration only. Exclude the pseudo-query’s source row from any reference-conditioned training/validation candidate set.
4. Freeze configuration, source hashes, seed list, fallbacks, brackets, and method registry before opening locked queries. Keep learned seeds separate in raw evidence before ensembling.
5. Save prediction coordinates or VT sets, scoring truth, physical block keys, runtime, feature provenance, map-information bracket, checkpoints, and hashes. Reconstruct every reported error mechanically from raw rows.
6. Run empty/no-evidence, padding, path-shuffle, candidate-shuffle, and finite-output tests. Cluster paired intervals by physical query/AP/scene, not augmentation row.
7. Label every number as paper context, full reconstruction, diagnostic smoke, oracle sensitivity, or pending. Never use a completed task’s number to fill another task’s missing table.

8 Bottom line

On the completed MAP replacement, the best dependable row is the same-acquisition range/power/AoA Beta-shaped survival adapter at 0.2528 m mean, versus 1.0638 m for corrected prose-consistent Majid. Survival-CIR and the multimodal residual graph are close; permutation-invariant self-attention and genuine cross-attention are also strong. Controlled feature ablations show that most of this margin comes from richer noisy sensing, not attention alone.

On SMART, the dominant method depends sharply on map density and receiver noise. Fixed-height geometric LEA4 is exceptionally accurate on dense clean maps but has a rare severe tail at 0.5 m; multimodal assignment dominates sparser maps, delay-CIR wins the densest noisy map, and multimodal attention wins noisy 0.5 m. The all-cell table must accompany any headline.

Original MPUrg VT construction and corrected CNN performance remain open release items. The report supplies exact recipes and input hooks but does not convert incomplete runs into claims. Separately, the causal analytic G2 filter is a confirmed positive sequential extension, whereas honest active multilevel calibration is neutral under paired uncertainty.